

Neural Ranging Inertial Odometry

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Background

UWB Localization:

- UWB is effective in GPS-denied localization for its scalability, affordable cost and laudable capacity to penetrate occlusion and provide drift-free location
- Challenges: sensor arrangement; multi-path; non-line-of-sight(NLOS);non-Gaussian pattern; multi-signal interference

Limitations of UWB-IMU Localization:

- Ignoring higher-order nonlinear terms
- Underutilization of robotic motion status
- UWB non-Gaussian pattern
- Unstable initial values
- Particle degradation or high computational cost

Motivation

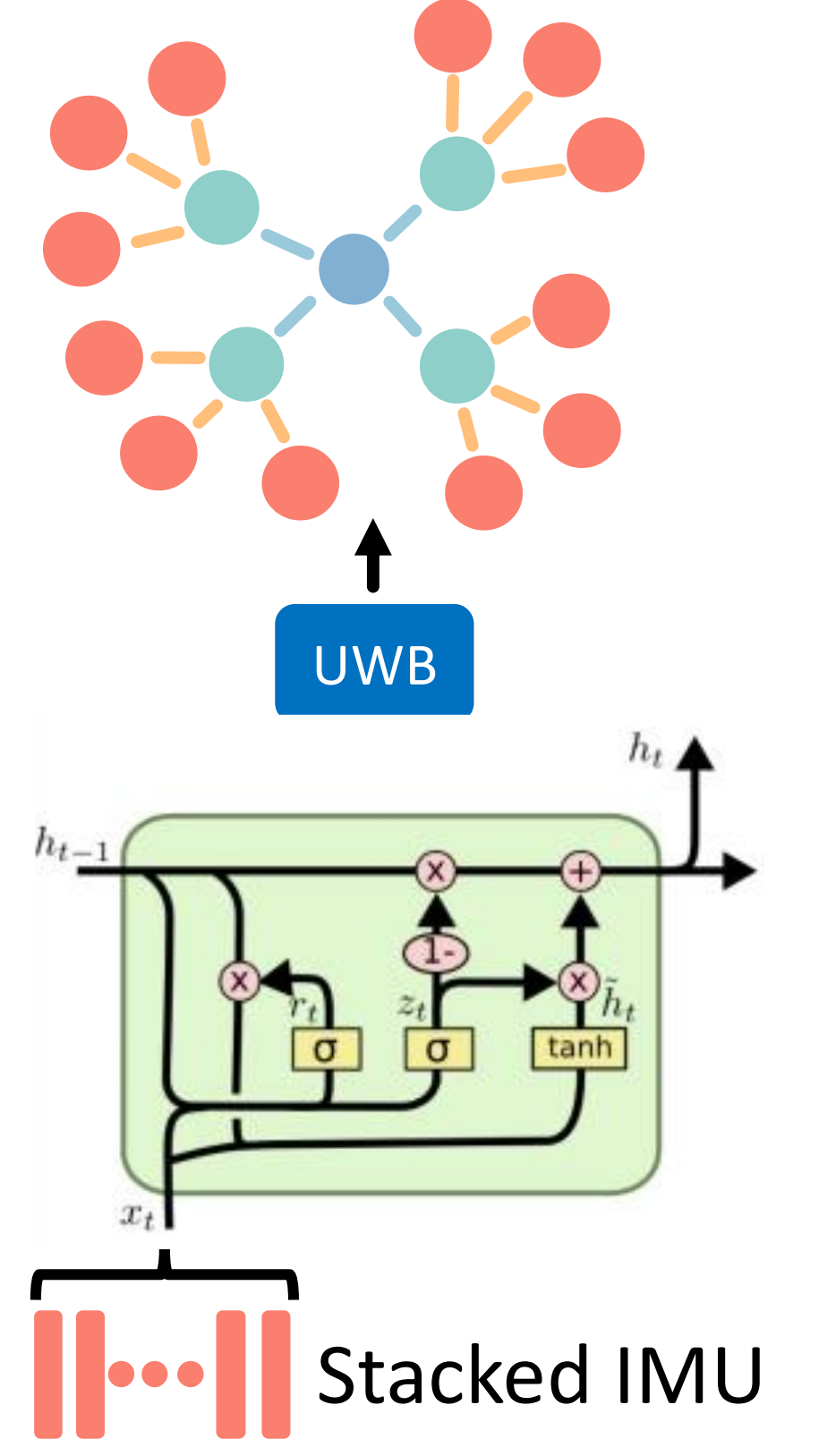
Graph Neural Network (GNN)

- Dynamism
- Aggregation power
- ✓ Adapt to any number of anchors or tags
- ✓ Model non-Gaussian patterns

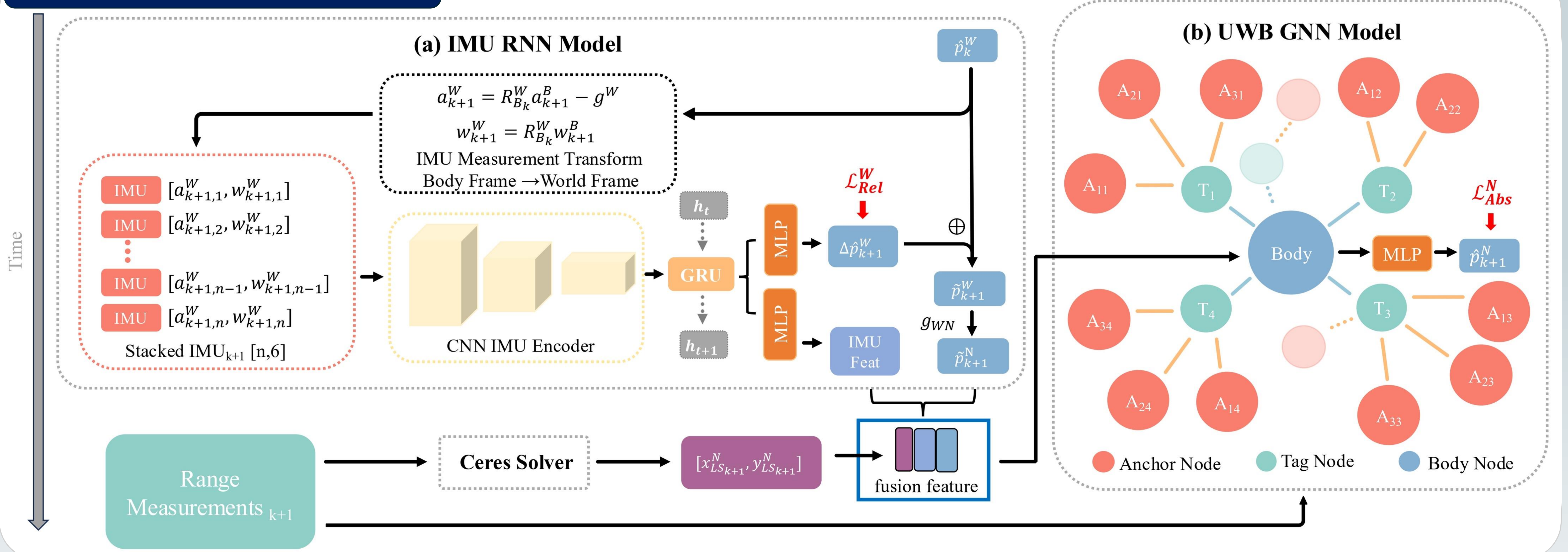
Recurrent Neural Network (RNN)

- Temporal features
- Robot motion features
- ✓ Neural IMU propagation

- We propose an IMU-RNN UWB-LS-GNN (IR-ULSG) model.



Frameworks



Experiment Result

